

Challenge (Habanero)

Q1. What is the vertex of the graph of $y = |x - 3| + 2$?

- (A) (3, 2)
- (B) (2, 3)
- (C) (3, -2)
- (D) (-3, 2)
- (E) (0, 3)

Q2. A group of hikers are planning a trip to a mountain. The cost of the trip is $|x - 10| + 50$ dollars, where x is the number of hikers. If 12 hikers are going, what is the cost?

- (A) 60
- (A) 70
- (A) 62
- (A) 52
- (A) 72

Q3. What is the axis of symmetry of the graph of $y = |2x + 1|$?

- (A) $x = -1/2$
- (B) $x = 1/2$
- (C) $x = 0$
- (D) $x = -1$
- (E) $x = 1$

Q4. The population of a certain animal species can be modeled by $y = |x - 5| - 3$. What is the minimum population?

- (A) -3
- (B) -2
- (C) -1
- (D) 0
- (E) 1

Q5. A car's fuel consumption is given by $y = |x - 40| + 10$ miles

per gallon, where x is the speed of the car in miles per hour. What is the most fuel-efficient speed?

- (A) 30 mph
- (B) 40 mph
- (C) 50 mph
- (D) 60 mph
- (E) 20 mph

Q6. What is the direction of opening of the graph of $y = -|x + 2| + 1$?

- (A) Up
- (B) Down
- (C) Left
- (D) Right
- (E) None

Q7. The graph of $y = |x| + 2$ is shifted 2 units to the left. What is the new equation?

- (A) $y = |x + 2| + 2$
- (B) $y = |x - 2| + 2$
- (C) $y = |x| + 4$
- (D) $y = |x| - 2$
- (E) $y = |x + 1| + 1$

Q8. What is the vertex of the graph of $y = |x + 1| - 3$?

- (A) (-1, -3)
- (B) (-1, 3)
- (C) (1, -3)
- (D) (1, 3)
- (E) (0, 0)

Open Ended Questions (FRQ)

Q9. Graph the function $y = |x - 2| + 1$. Identify the vertex, axis of symmetry, and direction of opening. Provide a brief explanation of each.

Q10. A park ranger is modeling the population of a certain animal species using the function $y = |x - 10| - 2$. If the population is currently at 5, what is the value of x ? Provide a step-by-step solution and explain the meaning of x in this context.

Chef's Tips

- Tip 1: Remind students to consider both positive and negative values when graphing absolute value functions.
- Tip 2: Encourage students to use graphing calculators or software to visualize the functions.
- Tip 3: Emphasize the importance of labeling the vertex, axis of symmetry, and direction of opening on the graph.